Reading the Room: Do LLM Tone Scores of Central-Bank Communication Carry Policy Signal? Evidence from Seven Central Banks in Three Languages

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Abstract

We ask whether hawkish/dovish tone scores produced by a small local large language model (LLM), `qwen3:4b` run through Ollama, carry measurable policy signal in the published decision documents of seven central banks (Fed, ECB, Bank of England, RBA, MAS, Bank Indonesia, Bank Negara Malaysia) in English, Indonesian and Malay. The study was pre-registered before any scorer touched the full corpus; every LLM result is re-estimated on masked text and on a post-training-cutoff split. **Every testable hypothesis is refuted at the pre-registered thresholds.** LLM tone does not improve out-of-sample prediction of the next rate decision over a persistence baseline at any of 4 adequately powered banks (closest: ECB, ΔLL 0.058, Holm p 0.13). It explains the announcement-window market move in 1 of 11 bank–market cells and beats a cited dictionary in none. Native-language and English versions of the same statement agree only moderately (Spearman ρ 0.61 for BI and 0.53 for BNM, against a pre-registered 0.70), although they almost never point in opposite directions. Scores move with the prompt wording (Krippendorff's α 0.65 at document level), and on the human-labelled FOMC test split the LLM reaches a macro-F1 of 0.48, above the dictionary (0.27) but below the pre-registered 0.50. The transfer hypothesis is untestable because a gated reference model could not be run. We release the corpus pipeline, scorers and evaluation harness as the Python package `cbtone`.

1. Introduction

Central banks publish a decision statement, and often minutes and a press conference, after every policy meeting. Turning that text into a number, a *tone* from dovish to hawkish, is now routine in research and in markets. The measurement literature, however, is overwhelmingly about the Federal Reserve and the ECB, almost entirely in English, and rarely asks whether the newest tool, a large language model (LLM) prompted to read the text, has *seen the outcomes it is asked to predict* during training.

This paper does three things. First, it releases `cbtone`, a Python library that fetches public central-bank communications politely, versions them as a hashed corpus, and scores them with four scorer families (a cited hawkish/dovish dictionary, a finance-tuned transformer, a local ≤ 4B LLM via Ollama, and a small fine-tuned multilingual model) behind one interface. Second, it runs a **pre-registered** study (tag `prereg-v1`, frozen before any scorer touched the full corpus) on seven headline central banks, the Fed, ECB, Bank of England, Reserve Bank of Australia, Monetary Authority of Singapore (MAS), Bank Indonesia (BI) and Bank Negara Malaysia (BNM), in English, Indonesian and Malay. Third, it treats training-data contamination as a first-class threat: every LLM result is re-estimated on **masked** text (bank, committee and instrument names, countries, dates and rate levels replaced by placeholders) and on a **post-cutoff** split, and a finding is called *robust* only if it survives both.

The five research questions ask whether tone predicts the next decision (RQ1), explains the market reaction to the announcement (RQ2), is consistent between the native-language and English versions of the same statement (RQ3), is stable across prompts, temperatures and model tags and agrees with human labels (RQ4), and transfers from the

Fed to Southeast Asia (RQ5). Every threshold was fixed in advance; results that contradict a hypothesis are reported as refuted.

2. Related work

Measurement moved from counting words to classifying them, mostly for the Fed. Dictionary approaches count hawkish and dovish terms (;) or adapt finance lexicons (;). Topic models of deliberation (;) and voice-tone measures () widen the lens. Supervised transformers came next: FOMC-RoBERTa, trained on 2,480 hand-labelled FOMC sentences, reaches a weighted F1 of 0.72 against about 0.50 for a rule-based baseline and 0.59 for zero-shot ChatGPT-3.5 (); CentralBankRoBERTa () classifies economic sentiment rather than policy stance. Zero-shot LLM reading of FOMC stance is modest (); a multi-bank study finds a fine-tuned multilingual encoder far ahead of GPT-4o (); and prompt format alone can move accuracy by tens of points ().

Contamination is measurable: LLMs recall macro outcomes almost perfectly before their training cutoff and near chance after it (); masking entity names changes LLM sentiment signals (); formal look-ahead tests exist (). No study we found combines masking and a post-cutoff split on central-bank text.

The event-study template is standard (; ; ;). Text-based measures explain far less of daily moves than rate surprises; changes in statement wording reach pseudo-R^2 up to 0.12 (). Evidence for Southeast Asia is thin: a Malaysian yield-curve event study (), regional transparency surveys (), and a description of Singapore's exchange-rate regime (); we found no study linking MAS statement text to the Singapore dollar, and none comparing tone between native-language and English versions of the same central-bank document.

3. Data

Corpus. Adapters fetch each bank's public pages under robots.txt, a ≥ 2 s per-host rate limit, and a descriptive User-Agent, caching every response with its SHA-256. MAS serves a bot page to automated clients and pbc.gov.cn disallows crawlers; with the author's approval, both are read from Internet Archive copies, never from the banks' own sites. The snapshot used here (`2026-10-14+ed5a49c28f7b41acaf4f21bda0d43570bd23049b363717c08d5efb3b9177bb8b`) holds 2,935 documents. Each bank's primary decision series is fixed in the pre-registration: FOMC statements, ECB decision press releases, BoE MPC minutes (its 2000–2006 releases are a single sentence), RBA statements from December 2007 (earlier releases exist only when the rate changed), MAS Monetary Policy Statements, and BI and BNM statements (English primary; Indonesian and Malay twins for RQ3). Text is committed only where the source's terms allow redistribution (Fed, ECB decision releases, RBA); for the other banks the repository holds URLs and hashes, and `make fetch` rebuilds the text locally.

Decisions. Rate decisions come from the BIS central-bank policy-rate series, matched to meeting dates from the corpus; holds are meetings without a change. MAS sets an exchange-rate band, not a rate, so its decisions (tighten / maintain / ease) come from MAS's own published decision table.

Markets. Daily FX closes (Yahoo Finance; bad ticks filtered by a pre-specified rule) and 2-year government yields where a free daily official source exists (US, euro area, UK, Australia). Moves are taken over $[t-1, t+1]$ closes around the release; a document without a release time is timed at 00:00 local on its release date, so the window always contains it.

Labels. Human labels exist only for the Fed: the FOMC hawkish/dovish test split (), used for evaluation only under its CC-BY-NC licence. For Southeast Asia, 200 sentences (MAS 60; BI 35 en + 35 id; BNM 35 en + 35 ms) carry **silver** labels from two LLM passes with different prompts and a tie-break; they are never called gold.

4. Method

Scorers. All scorers return probabilities over {hawkish, dovish, neutral} and an index *hawkish* − *dovish* ∈ [−1, 1], computed on the same input: the decision, forward-guidance, outlook and risk paragraphs of a document (rule-based tagger), capped at 6,000 characters. The *dictionary* uses a 167-term cited lexicon with negation handling (English only). The *LLM* is `qwen3:4b` run locally through Ollama at temperature 0 with a registered prompt (P1) and structured JSON output; all calls are cached by content hash. The *fine-tuned* model is XLM-R base trained for ≤ 30 minutes on the LLM's own sentence labels (never on FOMC data). FOMC-RoBERTa is implemented but is a gated model whose access was pending at the time of the run; it is reported as not run.

RQ1. Per bank, an ordered logit (cut < hold < hike) on (tone, Δtone, last direction) is refit at every decision on an expanding window (initial 40 %, at least 30) and predicts the next decision. The baseline is persistence (smoothed P(next | last)). The gain ΔLL = LL(reference) − LL(model) in multiclass log-loss is tested with a moving-block bootstrap (block 4); Holm correction across adequately powered banks; and the real gain must beat shuffled-date and permuted-label placebos (1,000 permutations each).

RQ2. The tone surprise is tone minus its AR(1) forecast fitted on earlier documents only. Signed FX (positive = domestic appreciation) and 2-year-yield moves are regressed on it with Newey–West errors; the R^2 must beat a non-announcement-day placebo, and the LLM's R^2 is compared with the dictionary's by paired block bootstrap.

RQ3–RQ5. Cross-lingual consistency uses Spearman ρ and the stance-flip rate on parallel documents. Stability uses Krippendorff's α and ICC(2,1) across five prompt paraphrases, two temperatures, two model tags and three aggregation strategies. Agreement with human labels is macro-F1 on the FOMC test split. Transfer compares each Fed-calibrated scorer's FOMC macro-F1 with its silver-set macro-F1 per bank.

Power and multiplicity. Each (bank, RQ) cell has a pre-set minimum sample; underpowered cells are reported in the appendix and excluded from Holm families and majority counts, and a majority needs at least three powered banks.

Compute. All LLM scores come from one backend: the same `qwen3:4b` weights (digest `359d7dd4`) served by Ollama on a desktop RTX 4070 Ti, where the full pre-registered workload fits the 2-hour LLM budget without any step of the pre-registered compute ladder. On the MacBook Air M4 named as the project's compute target, the same model processes about 220 prompt tokens/s (≈ 3.5 s per document at a 3,000-character cap), so a laptop run needs the ladder (Section 9).

5. Results

5.1 RQ1: does tone predict the next decision?

H1 is refuted. 4 banks meet the pre-registered power rule (≥ 40 out-of-sample predictions with ≥ 8 hikes and ≥ 8 cuts): Fed, ECB, RBA and BI. The LLM succeeds at none of them (Table 1, Figure 1).

Table 1: Table 1. RQ1, LLM tone, adequately powered banks. ΔLL = LL(reference) − LL(model), multiclass log-loss per decision; positive is better. Holm correction across the four banks.

Bank	OOS *n*	ΔLL vs persistence [95% CI]	Holm *p*	ΔLL vs dictionary	Beats placebos
Fed	135	−0.005 [−0.059, 0.039]	1.00	0.011	no
ECB	178	0.058 [0.007, 0.112]	0.13	0.052	yes
RBA	118	0.020 [−0.010, 0.050]	0.56	−0.023	no
BI	76	0.016 [−0.041, 0.083]	1.00	−0.016	no

The ECB is the only bank where the LLM's gain over persistence has a bootstrap interval above zero and exceeds both placebo distributions (95th percentiles 0.014 for shuffled tone and 0.010 for permuted labels); it does not survive the Holm correction. At the RBA the dictionary, not the LLM, has an interval above zero (ΔLL 0.043 [0.002, 0.087]). BoE (only 5 out-of-sample hikes), MAS (29 predictions) and BNM (7 hikes) are underpowered: they are marked in Figure 1 but excluded from the test. Ranking metrics are high for every scorer, including persistence (Fed persistence AUROC 0.77), because most of the information about the next move is already in the last one. The masked LLM gives nearly the same ECB estimate (0.059); BI's masked AUROC of exactly 1.00 rests on 20 hike/cut cases and is discussed in `ANOMALIES.md` (A9).

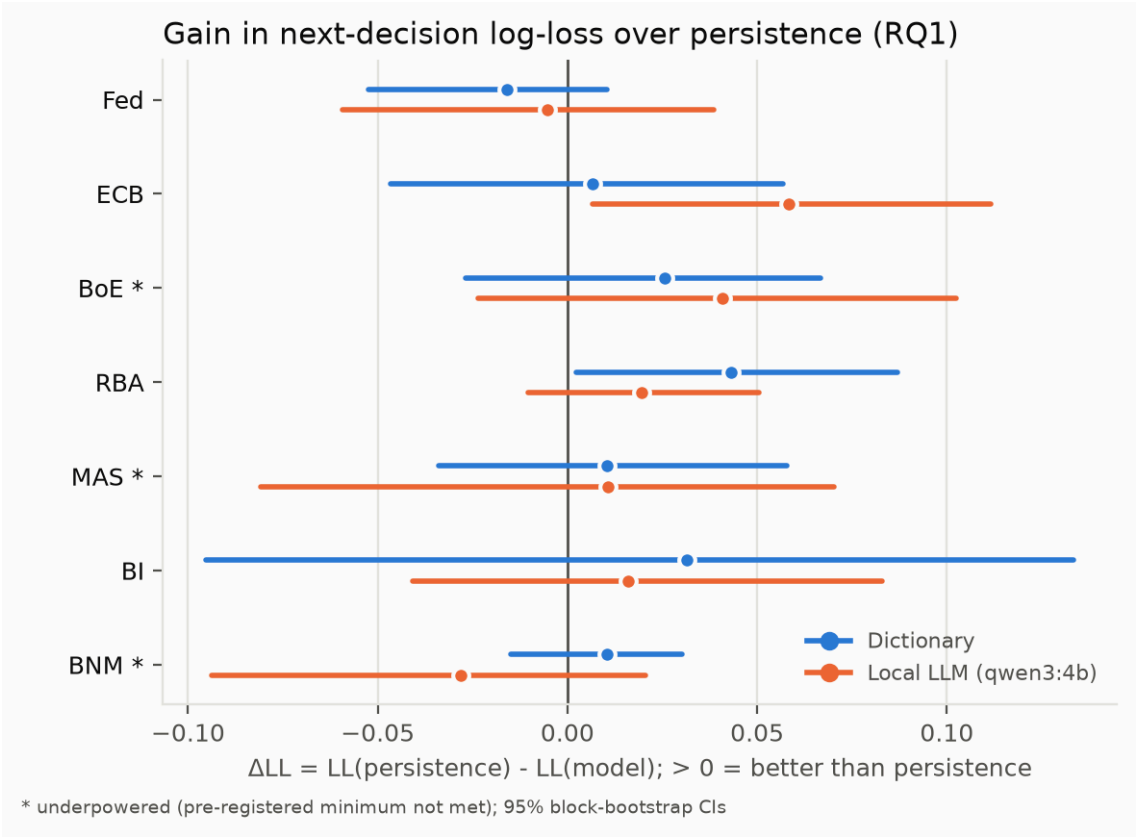


Figure 1: Figure 1. RQ1: gain in next-decision log-loss over persistence, by bank and scorer. * = underpowered.

5.2 RQ2: does the tone surprise explain the market reaction?

H2a and H2b are refuted. Of 11 adequately powered bank–market cells, the LLM tone surprise explains the release-window move beyond its placebo in 1 (Table 2), and its R^2 beats the dictionary's, after Holm correction, in 0.

Table 2: Table 2. RQ2, LLM tone surprise, all powered cells. FX β is the signed two-day log move (positive = domestic appreciation) per unit surprise; yield β is in basis points per unit surprise. Newey–West standard errors.

Bank	Series	*n*	β	R^2	Placebo R^2 p95	Holm *p* (β)	ΔR^2 vs dictionary
Fed	USD index	218	0.0016	0.003	0.018	1.00	0.003
Fed	US 2Y	218	3.25	0.014	0.019	0.40	0.008
ECB	EUR/USD	228	−0.0007	0.001	0.017	1.00	−0.001
ECB	EA 2Y	218	0.51	0.001	0.018	1.00	0.001
BoE	GBP/USD	235	0.0005	0.000	0.018	1.00	0.000
BoE	UK 2Y	275	1.00	0.002	0.016	1.00	0.002
RBA	AUD/USD	192	0.0041	0.012	0.019	0.83	0.012
RBA	AU 2Y	137	9.04	0.100	0.029	0.04	0.088
MAS	USD/SGD	53	0.0022	0.029	0.074	0.92	0.004
BI	USD/IDR	120	−0.0039	0.064	0.031	1.00	0.046
BNM	USD/MYR	133	0.0043	0.027	0.029	0.08	0.025

The single success is the RBA two-year yield: a one-unit hawkish surprise goes with a 9.0 bp rise (R^2 0.10, placebo p95 0.029). Its R^2 gain over the dictionary has a bootstrap interval above zero ([0.001, 0.225]) but a Holm-adjusted *p* of 0.43. For BI the LLM coefficient is significant with the opposite sign to the hypothesis (β -0.0039, unadjusted *p* 0.007):

more hawkish surprises coincide with rupiah depreciation over the day-level window. The one-sided test does not count it, and the design cannot separate reverse timing from any other explanation (A10). Pooled across banks with bank fixed effects, the FX coefficient is 0.0005 [−0.0010, 0.0020] (n = 1179).

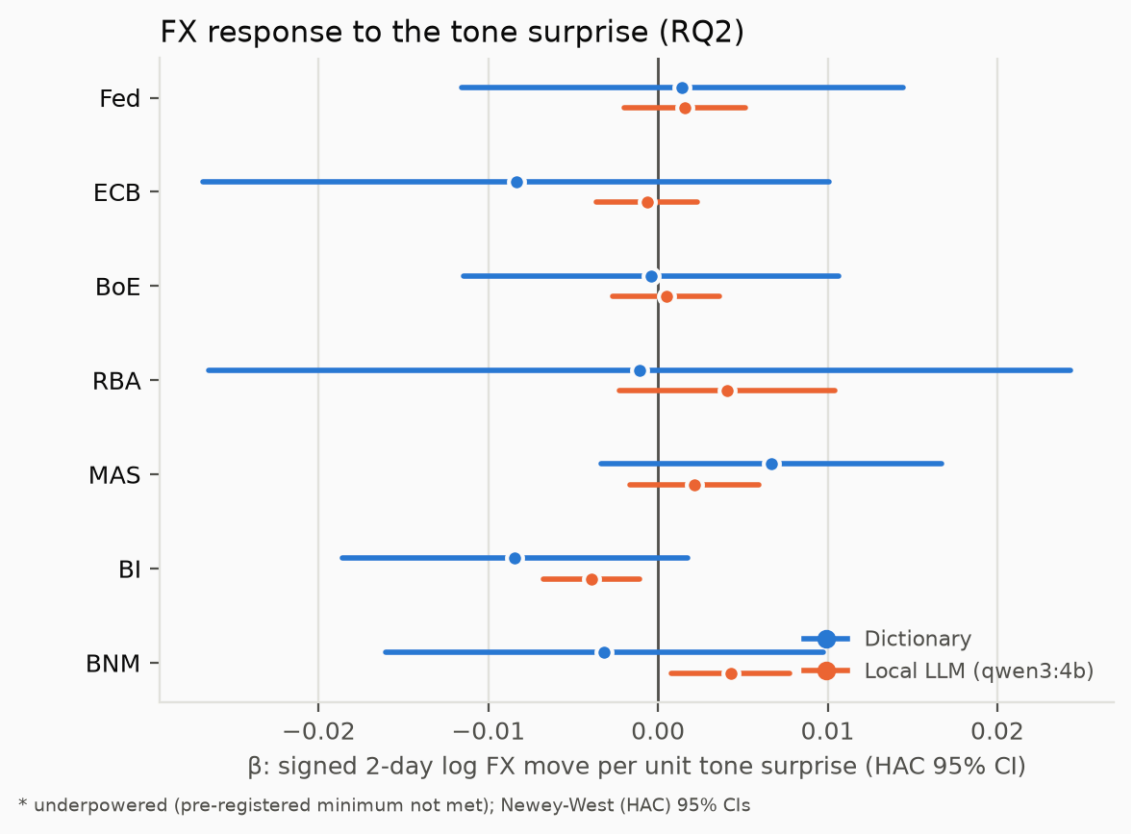


Figure 2: Figure 2. RQ2: FX response to the tone surprise, by bank and scorer (HAC 95% CIs).

5.3 RQ3: do native-language and English versions agree?

H3 is refuted. On parallel statements the LLM's native-language and English scores have Spearman ρ = 0.61 for BI (Indonesian, n = 128) and 0.53 for BNM (Malay, n = 139), below the pre-registered 0.70. Disagreement is about intensity, not direction: the stance flips in 2.3% and 2.2% of pairs, well under the 10 % ceiling, while the mean absolute gap is 0.28 and 0.18 on a [−1, 1] scale (Figure 3). The fine-tuned multilingual model is no more consistent (ρ 0.44 and 0.54). The dictionary is English-only and therefore not applicable; PBoC has 9 parallel pairs and is exploratory.

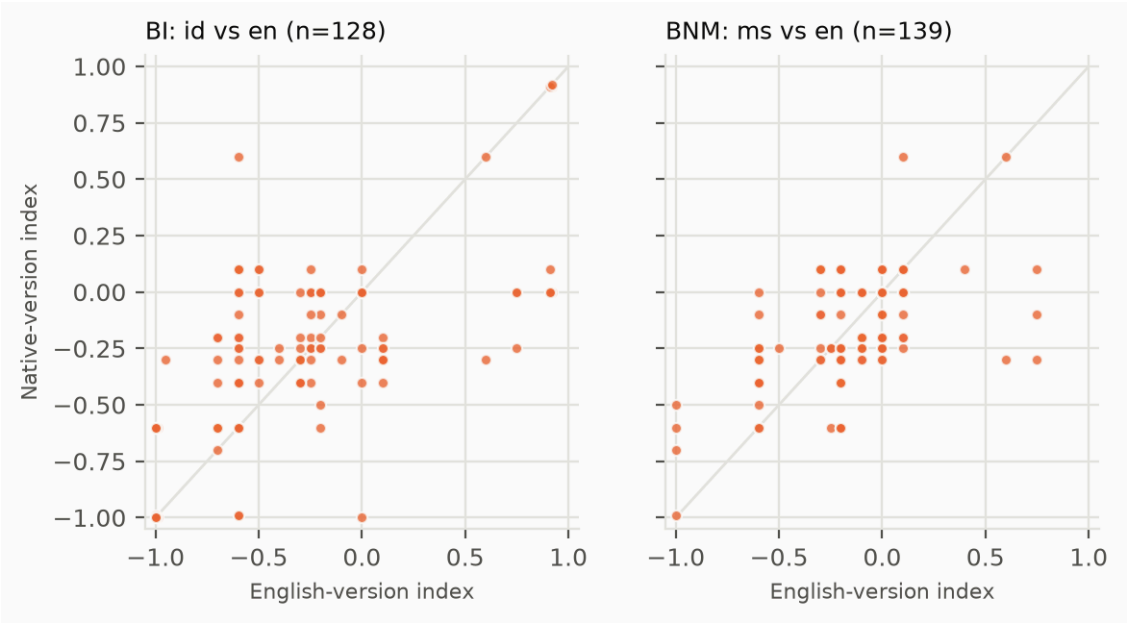


Figure 3: Figure 3. RQ3: LLM tone of the native-language version against the English version of the same statement.

6. Contamination and reliability

6.1 Stability (H4a: refuted)

Across the five registered prompt paraphrases, document scores reach Krippendorff's α 0.65 and ICC(2,1) 0.66 on 120 documents, short of the pre-registered 0.667 and 0.75; on 1000 sentences α is 0.42. Temperature matters much less (α 0.92 for documents, 0.81 for sentences, at 0 vs 0.7). Swapping the model tag for `qwen2.5:3b` changes scores most: α 0.46 for documents and −0.03 for sentences, i.e. no better than chance agreement at sentence level. Aggregating sentence scores into a document score gives indices that correlate with the direct document score at ρ 0.62–0.72, so not every pair of strategies clears 0.70. A tone index from a small local model is therefore partly a property of the prompt and the model version, which any user of such scores should report.

6.2 Agreement with human labels (H4b: refuted)

On the 496-sentence FOMC test split the LLM's three-class macro-F1 is 0.48 [0.43, 0.53]. It exceeds the dictionary by 0.21 (CI [0.16, 0.26]) but misses the pre-registered floor of 0.50, so H4b is refuted. The fine-tuned XLM-R, trained only on the LLM's labels for corpus sentences (no FOMC data), reaches 0.41. Our dictionary scores 0.27, well below the ≈ 0.50 reported for a rule-based baseline by Shah et al. (); the two rule sets differ, and ours returns neutral whenever hawkish and dovish matches balance or none appear, which is the case for almost every FOMC test sentence. The LLM also over-predicts the neutral class relative to the human labels. FOMC-RoBERTa, the in-distribution reference (≈ 0.72), could not be run (gated model).

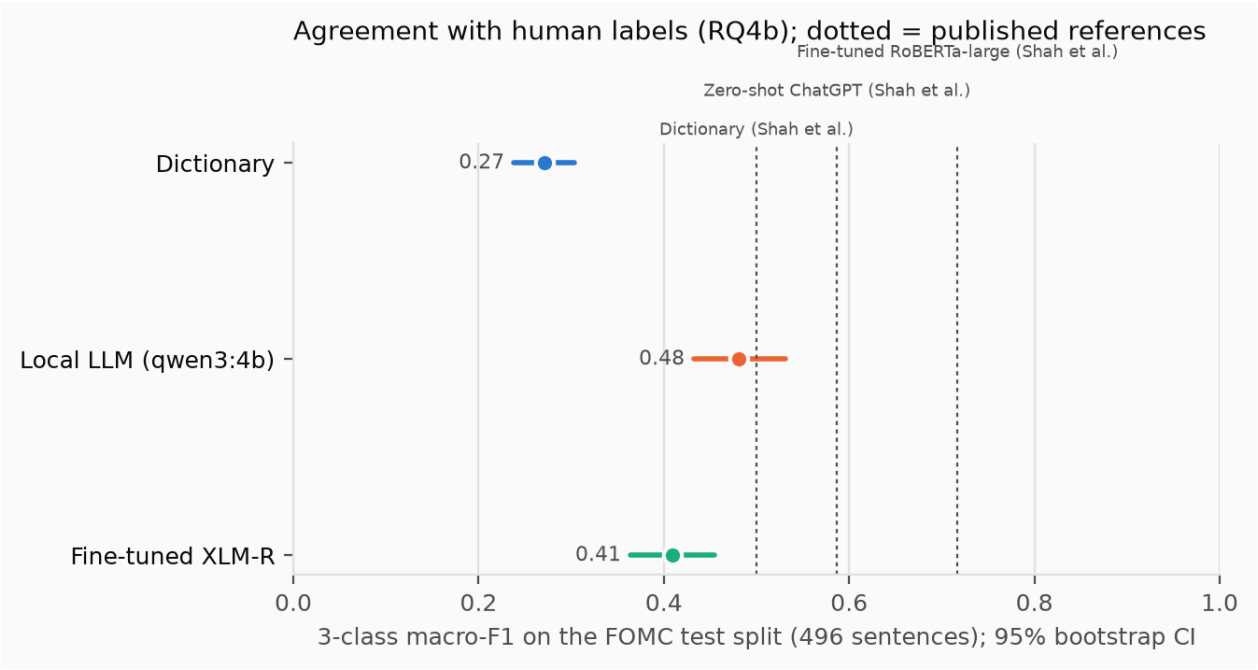


Figure 4: Figure 4. RQ4b: macro-F1 on the FOMC test split, with published reference values (dotted).

6.3 Contamination (H4c)

No RQ1 or RQ2 finding is robust. Pooled across powered banks, the LLM's log-loss gain over the dictionary is 0.007 [−0.016, 0.031] on original text and 0.004 [−0.017, 0.028] on masked text: there is no effect to preserve, so the masked check cannot pass. The post-cutoff window (after 2025-04-29) holds 58 decisions and 78 FX events; its estimates overlap the full-sample intervals but are too wide to be informative. Masking leaves the ECB point estimate nearly unchanged, which is consistent with, but does not prove, the absence of outcome memorisation for that series.

7. Transfer to Southeast Asia

H5 is untestable as pre-registered. It requires FOMC-RoBERTa, which could not be run, and the BI-en and BNM-en silver subsets have 35 sentences each, under the minimum of 50. Descriptively, on the 60 MAS silver sentences the dictionary's macro-F1 is 0.36 against 0.27 on FOMC sentences, so it does not lose performance on MAS (drop −0.09). The LLM and the fine-tuned model score higher on silver than on FOMC labels (0.64 and 0.67 on MAS), but the silver labels come from the same model family, so these numbers measure self-agreement rather than accuracy and support no claim. The two silver passes agree on 71% of sentences. In a multilingual sentence-embedding space, MAS sentences are the farthest from Fed sentences (mean cosine distance 0.58 vs 0.54 for the ECB), consistent with MAS's exchange-rate framing.

8. Threats to validity

- **Small samples for Southeast Asian banks.** BI text starts in April 2016 (the bank's index lists nothing older) and BNM changes its rate rarely, so several cells fall below the pre-registered minimum and are reported as underpowered rather than tested.
- **Silver labels.** The Southeast Asian evaluation set is labelled by the same model family as the LLM scorer; the LLM's own silver score is therefore reported but not used for any hypothesis.
- **Model cutoff.** The post-cutoff window (after the model's public release date, 29 April 2025) is short, so the post-cutoff check is often inconclusive; an inconclusive check means a finding is *not established as robust*.
- **Document selection.** One primary series per bank; BoE uses minutes because its early releases carry no text; RBA starts in December 2007.
- **Timing.** MAS and BI publish no release time; their event windows span the whole release day.
- **MAS decisions.** Taken from MAS's own table; the rule-based extractor's accuracy is development accuracy only and is used for two 2026 statements, which are excluded from tests.
- **Gated scorer.** FOMC-RoBERTa could not be run, which makes the transfer hypothesis (H5) untestable as pre-registered.

- **Compute.** The local 4B model is far smaller than hosted LLMs; results speak to what an analyst can run on local hardware, not to frontier models. The headline run used a desktop GPU; a laptop gives the same model and settings, more slowly.

9. Reproducibility statement

All code is Apache-2.0 at . `make reproduce` rebuilds every table and figure from the committed corpus snapshot (scores are cached by content hash, so a re-run repeats only the few LLM calls that failed the first time; the fine-tuned model is retrained, which on Apple MPS is not bit-for-bit deterministic; `ANOMALIES.md` A13). The pre-registration is the git tag `prereg-v1`; every later deviation is listed in `docs/DECISIONS.md` (D17–D20) and every data anomaly in `docs/ANOMALIES.md`. LLM scores were computed on a desktop RTX 4070 Ti (D20); the fine-tuned model was trained on an Apple M4 laptop (16 GB, MPS) in 6 minutes (`results/hardware.json`, `results/ft_train_report.json`); per-stage wall time is in `results/timing.json`.

10. Conclusion

A small LLM run on a consumer desktop GPU (or, more slowly, a laptop) produces tone indices that look plausible over time (Figure 5) and agree with human FOMC labels better than a simple dictionary, yet on every pre-registered test of *policy signal* (next-decision prediction, market reaction, cross-lingual consistency, prompt stability) the evidence is negative or underpowered. Three lessons follow for practitioners. First, persistence is a strong baseline for decision prediction, and gains from text are small and bank-specific. Second, LLM tone depends on the prompt and the model version, so a single-prompt index should not be treated as a measurement without a stability check. Third, for Southeast Asian banks the binding constraint is sample size: BI's text begins in 2016, BNM rarely moves, and MAS met only twice a year until 2024, so BNM and MAS are underpowered for decision prediction and the silver evaluation sets are small. The library, the masked and post-cutoff checks, and every number in this paper can be regenerated with `make reproduce`.

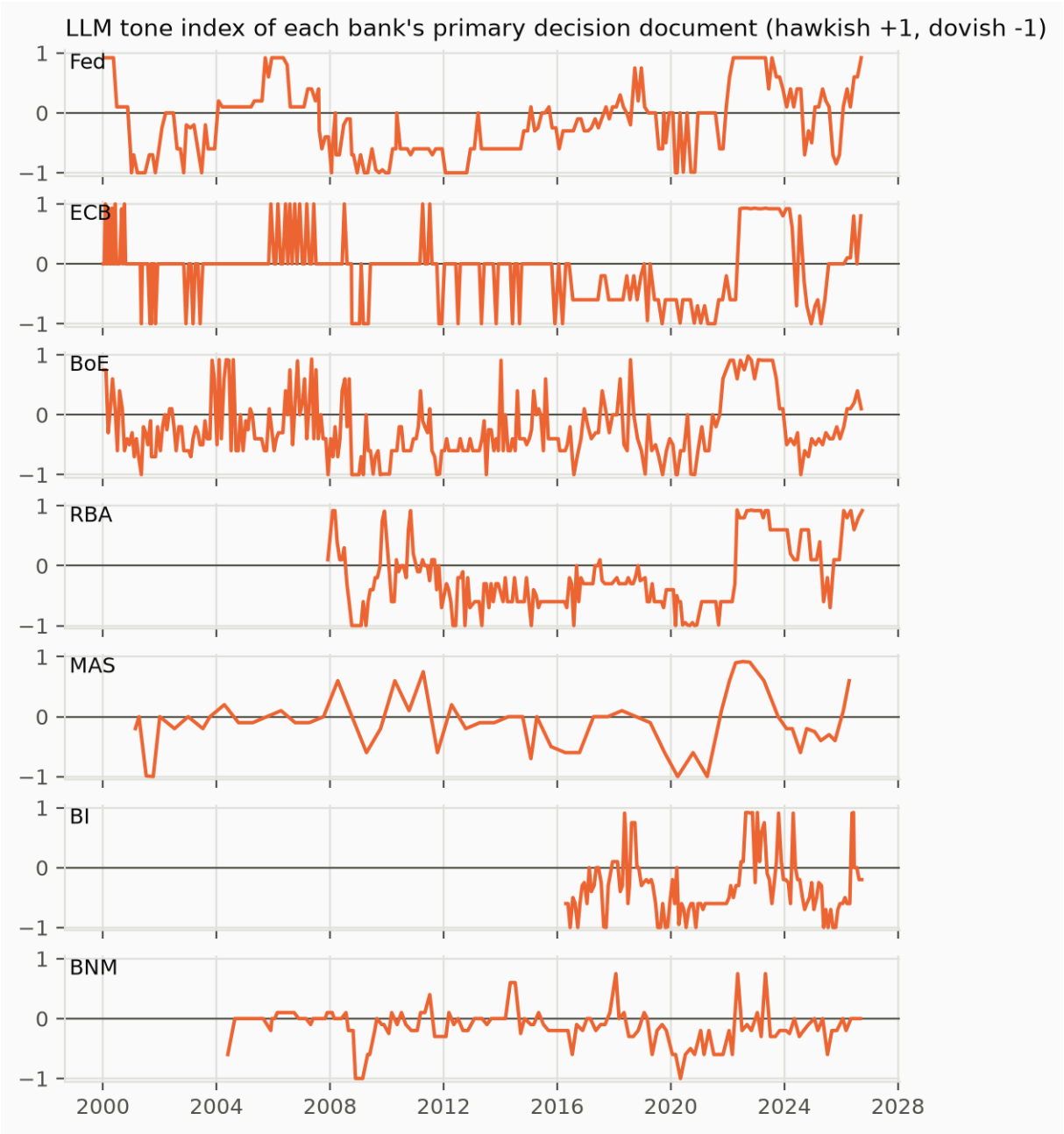


Figure 5: Figure 5. LLM tone index of each bank's primary decision document.

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